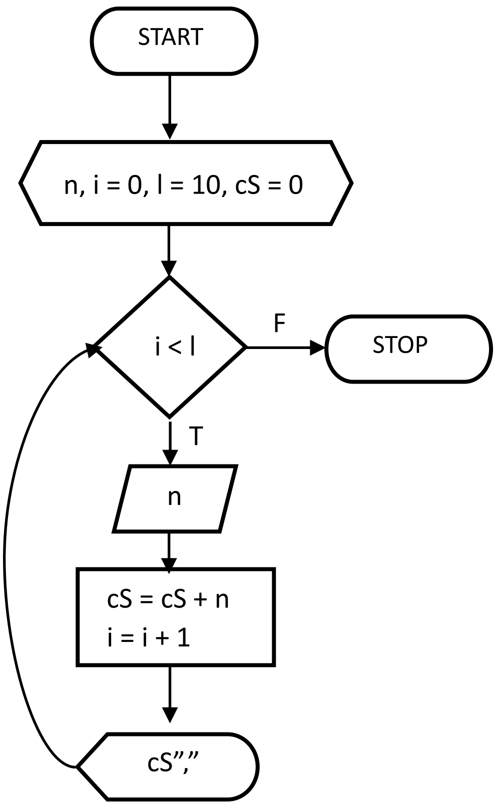
FOP Mentorship Module 1 Exercise

Flowcharts and Pseudocode: For flowchart- trace the output of the indicated flowchart and briefly explain. For pseudocode- write the appropriate statement, variable or constant in order to satisfy the indicated output.

<table>
<tr>
<td rowspan="2">

Output(Given that n is 2, 5, 6, 10, 50, 9, 2, 6, 10, 20):

Explanation:</td>
<td>Given a number, fill in the blanks the appropriate algorithm to count the digits of that number. E.g. input n = 6023562
Output: 7</td>
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<tr>
<td>ctr = 0
n
input ______
while(________)
| ____________________
| ____________________
end
display ctr</td>
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I. **Programming on Paper:** The area of a triangle is solved through the use of this equation:

$$A = \frac{1}{2}bh$$

where, b = base and h = height

Write a complete C code of the equation above and then display A for fixed values b = 5 and h = 10, b = 1 and h = 3, and b = 15, h = 30.